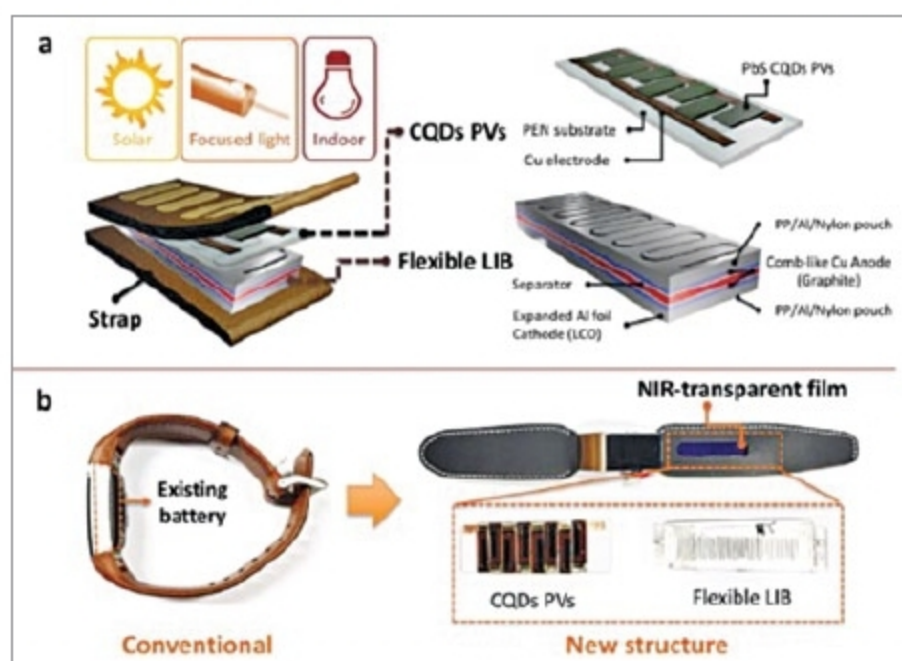


cal treatments for a variety of medical conditions in humans. This type of technology, which they refer to as bioresorbable electronic medicine, provides therapy and treatment over a clinically-relevant period of time and directly at the site where it is needed, thereby reducing the side effects or risks associated with conventional, permanent implants.

Wireless self-charging system using NIR band

A research team at Korea Advanced Institute of Science and Technology (KAIST), South Korea, has developed a permanent, wireless self-charging platform for low-power wearable electronics by converting near-infrared (NIR) band irradiation into electrical energy. This technology can be applied to flexible,



(a) Conceptual NIR-driven self-charging system including a flexible CQD PV module and an inter-digitatedly structured LIB (b) A conventional wearable healthcare bracelet and a self-charging system-integrated wearable device

wearable charging systems without any attachments.

Colloidal quantum dots (CQDs) are promising materials for manufacturing semiconductors. PbS-based CQDs have facile optical tunability from the visible to IR wavelength region. Hence, these can be applied to devices such as lighting, photovoltaics (PVs) and photodetectors.

The joint team led by professor Jung-Yong Lee from Graduate School of Energy, Environment, Water and Sustainability and Jang Wok Choi from Seoul National University decided to apply CQD PVs, which have high quantum efficiency in NIR band, to self-charging systems on wearable devices. The team employed a stable and efficient NIR energy conversion strategy. The system comprises PbS CQD-based PV module, flexible interdigitated lithium-ion battery and various types of NIR-transparent films.

Part-organic materials for bendable cellphones

Engineers at Australian National University, Australia, have created a semiconductor with organic and inorganic materials that

can convert electricity into light efficiently. The semiconductor is thin and flexible enough to help make devices such as cellphones bendable.

The organic component, which is one atom thick, is made from only carbon and hydrogen. It forms part of the semiconductor that the team has developed. The inorganic component is around two atoms thick. The hybrid structure can convert electricity into light efficiently for displays on cellphones, televisions and other electronic devices.

The team grew the organic semiconductor component molecule by molecule, in a process similar to 3D printing. The process is called chemical vapour deposition.

This invention also paves the way for a new generation of high-performance electronic devices made with organic materials that will be biodegradable or be easily recycled. This promises to help reduce e-waste substantially. Huge volumes of e-waste generated by discarded electronic devices around the world is causing irreversible damage to the environment.

Robotic skins turn everyday objects into robots

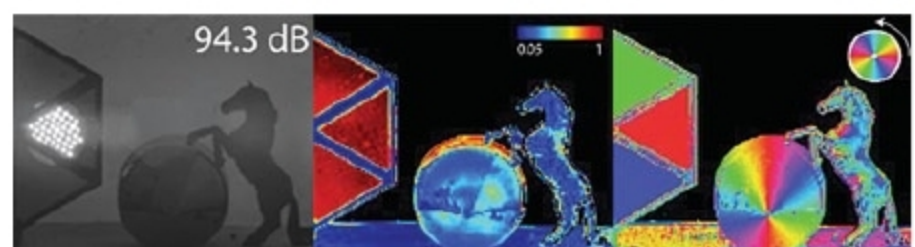
New robotic skins technology developed by Yale University, USA, researchers allows users to animate inanimate and everyday objects into robots. Developed in the lab of Rebecca Kramer-Bottiglio, assistant professor of mechanical engineering and materials science, these robotic skins could be used for everything from search-and-rescue robots to wearable technologies.

The robotic skins are made from elastic sheets embedded with sensors and actuators developed in Kramer-Bottiglio's lab. Placed on a deformable object, such as stuffed animal or foam tube, the skins animate these objects from their surfaces. The makeshift robots can then perform different tasks depending on the properties of soft objects and how skins are applied.

Bio-inspired camera to help self-driving cars see better

Inspired by the visual system of the mantis shrimp, researchers from University of Illinois at Urbana-Champaign, USA, have created a camera that could greatly improve the ability of cars to spot hazards in challenging imaging conditions.

The camera accomplishes this by detecting a property of light known as polarisation and featuring a dynamic range



Dynamic range and polarisation capability can be seen in the light intensity image (left) and two polarisation images (middle and right) acquired with the camera